

Telco Customer Churn dataset

Case 1

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Executive Summary

A subscription-based telecommunications company is experiencing stagnant revenue despite steady customer acquisition. Management suspects that customer churn is offsetting new growth but lacks clarity on which customers are most likely to leave and what factors drive their decisions. This analysis addresses that challenge by developing a predictive model to identify customers at high risk of churn and translate the results into actionable retention strategies.

Using the Telco Customer Churn dataset from Kaggle, which contains information on 7,043 customers, this study applies data cleaning, exploratory data analysis, and logistic regression modeling to predict customer churn. Key variables include customer tenure, contract type, internet service type, and billing characteristics.

The results show that churn is strongly associated with short customer tenure, month-to-month contracts, and fiber-optic internet service. Customers on two-year contracts are approximately 85% less likely to churn than those on month-to-month contracts, while fiber-optic customers are over three times more likely to leave. The final model demonstrates strong predictive performance, achieving an AUC of approximately 0.82, indicating good discrimination between churners and non-churners.

Based on these findings, this report recommends targeted retention strategies focused on early-tenure customers, incentives for contract upgrades, and proactive engagement of high-risk fiber-optic customers. By leveraging predictive analytics, management can move from reactive churn response to proactive, cost-effective retention.

1. Business Context and Problem Definition

The telecommunications industry is highly competitive, with customers facing low switching costs and numerous service alternatives. While customer acquisition is important, retaining existing customers is often more cost-effective than acquiring new ones. The business decision motivating this analysis is how to **identify customers at high risk of churn and intervene before they leave**.

The primary decision-makers are senior management and the customer retention team, who must allocate limited resources toward retention initiatives. This problem matters because unchecked churn erodes revenue, reduces customer lifetime value, and undermines long-term growth. A data-driven understanding of churn risk enables the company to prioritize retention efforts where they are most likely to yield returns.

2. Data Description and Preparation

2.1 Data Sources

This analysis uses the **Telco Customer Churn dataset** obtained from Kaggle, a public platform that provides real-world datasets for analytical and educational purposes. The dataset represents customer behavior in a subscription-based telecommunications company and is commonly used to study customer retention and churn prediction.

The dataset contains **7,043 customer records**, where each observation corresponds to an individual customer account. The **unit of analysis** is the customer, and the primary response variable is **Churn**, a binary indicator showing whether a customer has discontinued service (*Yes*) or remained active (*No*).

The dataset includes variables describing **customer demographics** (e.g., gender, senior citizen status), **account characteristics** (e.g., tenure, contract type), **service subscriptions** (e.g., internet service type, add-on services), and **billing information** (e.g., monthly charges, total charges, payment method). These variables provide a comprehensive view of customer behavior and service usage.

The data represent a **cross-sectional snapshot** of customers at different stages of their lifecycle rather than a time-series dataset. Customer identifiers are anonymized, and no sensitive personal information is included, making the dataset suitable for ethical analytical use. Overall, the dataset is well suited to support classification modeling and business-oriented churn analysis.

2.2 Data Cleaning and Processing

Data cleaning and processing are essential steps in any analytical study, as predictive models rely on accurate and well-structured data. Real-world business datasets often contain missing values, inconsistent data types, or variables that are not immediately suitable for modeling. If left unaddressed, these issues can negatively affect model performance and lead to misleading conclusions.

For a customer churn analysis, proper data preparation ensures that customer characteristics, service usage, and billing information are represented consistently and meaningfully. The preprocessing steps in this study were therefore designed to improve data quality while maintaining interpretability and preserving the full dataset as much as possible.

The specific data cleaning and feature engineering steps applied in this analysis are described below:

- **Missing data handling:**
The variable `TotalCharges` contained missing values due to non-numeric entries. These were converted to numeric, and missing values were imputed using the median to preserve sample size.
- **Data type corrections:**
Categorical variables were converted to factors, and the binary variable `SeniorCitizen` was recoded into interpretable categories (“Yes” / “No”).
- **Feature engineering:**
Customer tenure was grouped into meaningful categories (e.g., 0–12 months, 13–24 months) to capture lifecycle effects.
- **Assumptions:**
Median imputation assumes missing values are not systematically biased. Identifier variables (e.g., `customerID`) were excluded from modeling.

These decisions were guided by the need to balance data integrity with interpretability, rather than maximizing model complexity.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is used to examine the structure of the data and identify patterns relevant to customer churn before formal modeling. This step helps highlight important relationships between customer characteristics and churn behavior and informs variable selection for the predictive model.

The analysis shows that approximately **26.5% of customers have churned**, indicating a moderately imbalanced dataset. Customers who churn tend to have **shorter tenure**, with churn most common during the first year of service. Additionally, customers with **higher monthly charges** exhibit higher churn rates, particularly among those on **month-to-month contracts**. Contract length shows a strong association with churn, as customers on longer-term contracts churn at substantially lower rates than those without contractual commitment.

These findings provide initial evidence of key churn drivers and motivate the use of a classification model to quantify churn risk.

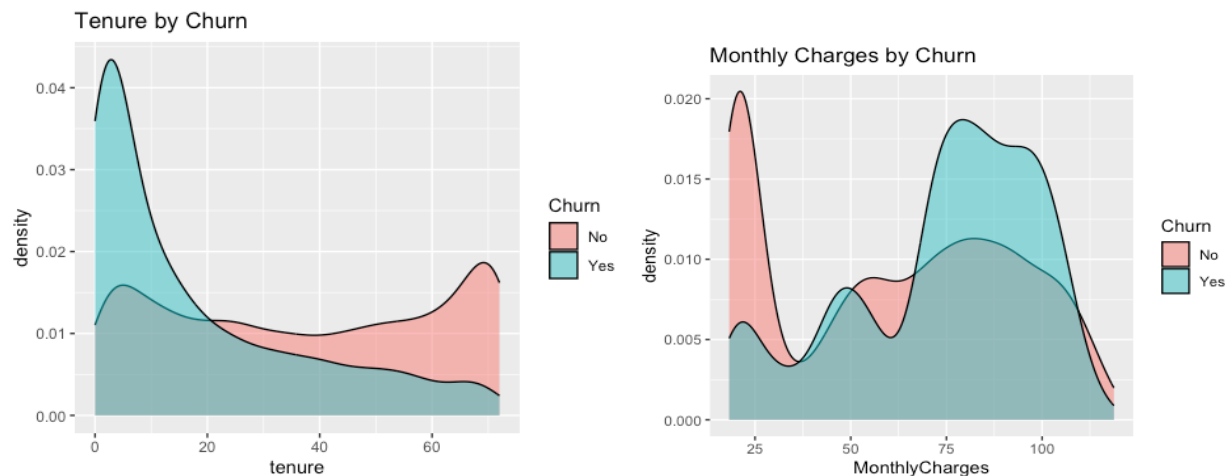


Figure 1 : Churn distribution

4. Methodology

4.1 Model or Analytical Approach

To predict customer churn, a **binary logistic regression model** was employed. Logistic regression is appropriate for this analysis because the response variable, **Churn**, is binary, indicating whether a customer has discontinued service (*Yes*) or remained active (*No*). The model estimates the probability of churn as a function of multiple customer and service-related characteristics.

The predictor variables include **customer tenure**, **monthly charges**, **total charges**, **contract type**, **internet service type**, and a **tenure group indicator**. These variables were selected based on patterns observed during exploratory data analysis and their relevance to business decision-making. Together, they capture customer lifecycle stage, pricing exposure, and service commitment, which are expected to influence churn behavior.

Logistic regression models the **log-odds of churn** as a linear combination of the predictor variables. This structure allows each coefficient to be interpreted as the marginal effect of a predictor on churn risk while holding other factors constant. From a business perspective, this interpretability is valuable because it enables management to understand not only which factors matter, but also the **direction and relative strength** of their impact on churn.

Alternative machine learning approaches, such as random forests and gradient boosting, were considered due to their predictive power. However, logistic regression was prioritized because it provides transparent, interpretable results that are well suited for communicating insights and supporting managerial decision-making.

4.2 Model Assumptions and Diagnostics

The logistic regression model relies on several key assumptions. First, observations are assumed to be independent, meaning that each customer's churn decision is not directly influenced by another customer's outcome. Given that each record represents a distinct customer account, this assumption is reasonable in the current context.

Second, logistic regression assumes a **linear relationship between the predictor variables and the log-odds of churn**. While this does not require a linear relationship between predictors and the outcome itself, it implies that changes in predictors have a consistent effect on churn risk on the log-odds scale.

Unlike linear regression, logistic regression does not require predictors to be normally distributed. However, the model may be sensitive to **multicollinearity** among predictors and to **omitted variables** that are not captured in the dataset. These limitations are acknowledged when interpreting the results, and findings are treated as associative rather than strictly causal.

5. Results

The logistic regression model identifies several predictors that are both **statistically significant and meaningful from a business perspective**. These results provide insight into which customer characteristics are most strongly associated with churn and help explain why certain customers are more likely to leave.

Contract length emerges as one of the most influential predictors of churn. Customers on **one-year and two-year contracts** are significantly less likely to churn compared to those on **month-to-month contracts**. In particular, customers on two-year contracts exhibit the lowest churn risk, indicating that longer contractual commitments play a strong role in customer retention.

Internet service type is also a key driver of churn behavior. Customers subscribed to **fiber-optic internet service** show a substantially higher likelihood of churn compared to customers using other internet services or no internet service. This suggests that fiber-optic customers may have higher service expectations or greater sensitivity to pricing and service quality.

Customer tenure shows a consistent negative relationship with churn. Customers who have been with the company longer are less likely to churn, while churn is concentrated among customers in the early stages of their relationship with the company. This finding highlights the importance of retaining customers during their first year of service.

Rather than focusing solely on coefficient values, these results are interpreted in terms of **relative risk and managerial relevance**. Together, they indicate that contract structure, service type, and customer lifecycle stage are central factors influencing churn and should be the primary focus of retention strategies.

6. Validation and Robustness Checks

Model performance was evaluated using a train-test split and multiple metrics. The model achieved an overall **accuracy of approximately 77.6%**, indicating solid predictive performance.

At the default probability threshold of 0.50, the model shows **high specificity ($\approx 91.3\%$)** but relatively low **sensitivity ($\approx 39.6\%$)**, meaning many churners would be missed. To address this limitation, the classification threshold was optimized using **Youden's Index**, increasing sensitivity to **over 80%** while maintaining reasonable specificity. The model's **AUC of approximately 0.82** further confirms strong overall discriminatory ability.

These results indicate that the model is robust and well suited to support data-driven customer retention decisions.

7. Managerial Recommendations

Based on the analysis, several targeted actions are recommended to reduce customer churn and improve retention.

First, the company should focus on **month-to-month customers** by offering incentives to move to **longer-term contracts**, such as discounted rates or bundled services, since contract length is the strongest predictor of churn. Second, **early-tenure customers** should receive enhanced onboarding and engagement during their first year, when churn risk is highest. Third, **fiber-optic customers** should be prioritized for proactive technical support and service quality monitoring due to their elevated churn risk.

Finally, the company should adopt **probability-based targeting** by using a lower churn threshold to identify high-risk customers earlier, allowing retention efforts to be applied more effectively. Together, these strategies support a proactive and data-driven approach to customer retention.

8. Conclusion

This case study demonstrates how predictive analytics can support decision-making in a subscription-based telecommunications business. By identifying key drivers of churn such as contract length, internet service type, and customer tenure, the analysis provides actionable insights for improving customer retention.

The results highlight the value of combining statistical modeling with business understanding to proactively reduce churn and support long-term profitability.

References

Kaggle. (n.d.). *Telco Customer Churn Dataset*.
Department of Statistics and Business Analytics. (2026). *STAT 482 Case 1: Customer Churn Prediction*.

Appendix

2.2, Data Cleaning and Processing:

2.2.1. Fix Data Types

```
df <- df %>%
  mutate(TotalCharges = as.numeric(TotalCharges))

sum(is.na(df$TotalCharges))

## [1] 11
```

2.2.2. Missing Value Handling

```
df <- df %>%
  mutate(TotalCharges = as.numeric(TotalCharges)) %>%
  mutate(TotalCharges = if_else(
    is.na(TotalCharges),
    median(TotalCharges, na.rm = TRUE),
    TotalCharges
  ))
```

2.2.3. Recoding

```
df <- df %>%
  mutate(
    across(where(is.character), as.factor),
    SeniorCitizen = factor(SeniorCitizen, levels = c(0,1), labels = c("No", "Yes")),
    Churn = factor(Churn, levels = c("No", "Yes"))
  ) %>%
  select(-customerID)
```

3, Exploratory Data Analysis (EDA)

```
prop.table(table(df$Churn))
```

```
##
##      No      Yes
## 0.7346301 0.2653699
```

```
ggplot(df, aes(x = tenure, fill = Churn)) +
  geom_density(alpha = 0.5) +
  labs(title = "Tenure by Churn")
```

```
ggplot(df, aes(x = MonthlyCharges, fill = Churn)) +
  geom_density(alpha = 0.5) +
  labs(title = "Monthly Charges by Churn")
```

4.1, Model or Analytical Approach

```
glm_fit <- glm(
  Churn ~ tenure + MonthlyCharges + TotalCharges +
    Contract + InternetService + tenure_group,
  data = train_df,
  family = binomial(link = "logit")
)
summary(glm_fit)
## Call:
## glm(formula = Churn ~ tenure + MonthlyCharges + TotalCharges +
##      Contract + InternetService + tenure_group, family = binomial(link = "logit"),
##      data = train_df)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.1118603   0.2250487  -0.497   0.6192
## tenure        -0.0542342   0.0107506  -5.045 4.54e-07 ***
## MonthlyCharges  0.0009615   0.0041036   0.234   0.8147
## TotalCharges   0.0001601   0.0000868   1.845   0.0651 .
## ContractOne year -0.8349944   0.1217922  -6.856 7.09e-12 ***
## ContractTwo year -1.8646980   0.2116066  -8.812 < 2e-16 ***
## InternetServiceFiber optic 1.1089214   0.1502548   7.380 1.58e-13 ***
## InternetServiceNo -0.9622809   0.1776636  -5.416 6.08e-08 ***
## tenure_group13-24 -0.2963519   0.1549252  -1.913   0.0558 .
## tenure_group25-48 -0.0853391   0.2719455  -0.314   0.7537
## tenure_group49-60  0.3995459   0.4348560   0.919   0.3582
## tenure_group61+   0.5080897   0.5402313   0.941   0.3470
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6113.6  on 5282  degrees of freedom
## Residual deviance: 4463.7  on 5271  degrees of freedom
## AIC: 4487.7
##
## Number of Fisher Scoring iterations: 6
```

5, Results

```
exp(coef(glm_fit))
##              (Intercept)              tenure
##              0.8941692              0.9472102
##              MonthlyCharges            TotalCharges
##              1.0009620              1.0001601
##              ContractOne year            ContractTwo year
##              0.4338769              0.1549430
## InternetServiceFiber optic            InternetServiceNo
##              3.0310872              0.3820206
##              tenure_group13-24            tenure_group25-48
##              0.7435257              0.9182009
##              tenure_group49-60            tenure_group61+
##              1.4911474              1.6621130
```